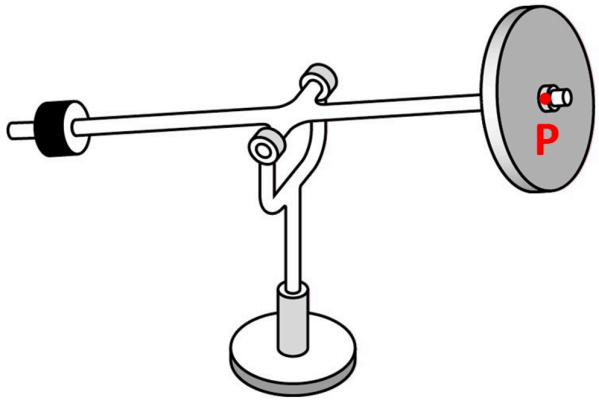
2P - Extra

Exercicis extra relacionats
amb angles d'Euler

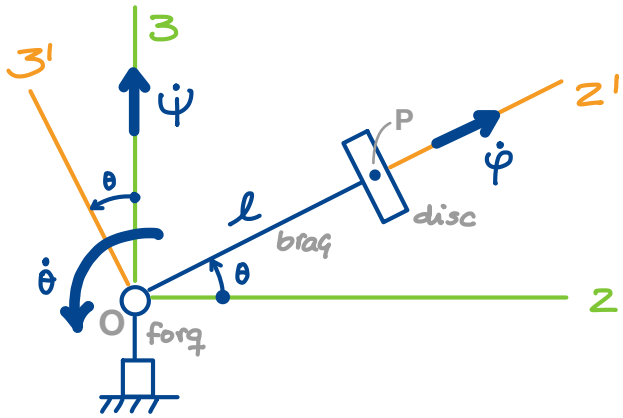
Versió 1.0



Tots els elements del giroscopi (suport, forquilla, braç, disc) estan articulats dos a dos. El suport és fix a terra (T).

Mitjançant la derivada analítica, calcula $\bar{v}_T(\mathbf{P})$, $\bar{a}_T(\mathbf{P})$, $\bar{\alpha}_T^{\text{braç}}$.

Definim les rotacions $\dot{\bar{\psi}}$, $\dot{\bar{\theta}}$ i $\dot{\bar{\varphi}}$ que s'indiquen al dibuix →



Les dues bases que van bé per treballar són:

(*)

$$\underbrace{B = (1, 2, 3)}_{\text{D'or. fixa a forq.}}$$

$$\bar{\Omega}_T^B = (\odot \dot{\theta})$$

(**)

$$\underbrace{B' = (1', 2', 3')}_{\text{D'or. fixa a braç}}$$

$$\bar{\Omega}_T^{B'} = (\odot \dot{\theta}) + (\Uparrow \dot{\psi})$$

Les dues bases van igual de bé. Per practicar, farem els càlculs en totes dues.

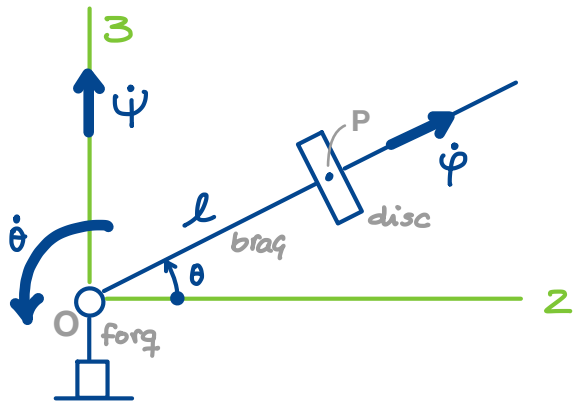
(*) Les dirs. 1 i 3 són les de $\dot{\bar{\theta}}$ i $\dot{\bar{\psi}}$, respectivament, i per construcció dels angles d'Euler són sempre $\perp$.

(**) Les dirs. 1' i 2' són les de $\dot{\bar{\theta}}$ i $\dot{\bar{\varphi}}$ (sempre $\perp$). A més:

$$\bar{\Omega}_T^{B'} = \bar{\Omega}_T^{\text{braç}} = \bar{\Omega}_{\text{forq}}^{\text{braç}} + \bar{\Omega}_T^{\text{forq}} = (\odot \dot{\theta}) + (\Uparrow \dot{\psi})$$

En base $B = (1,2,3)$

$\bar{v}_T(P)$ i $\bar{a}_T(P)$



$$\{\overline{OP}\}_B = \begin{Bmatrix} 0 \\ \ell\cos\theta \\ \ell\sin\theta \end{Bmatrix}$$

$\Big\downarrow$ derivant

$$\{\bar{v}_T(P)\}_B = \begin{Bmatrix} 0 \\ -\ell\dot{\theta}\sin\theta \\ \ell\dot{\theta}\cos\theta \end{Bmatrix} + \overbrace{\begin{Bmatrix} 0 \\ 0 \\ \dot{\psi} \end{Bmatrix}}^{\bar{\Omega}_T^B} \times \begin{Bmatrix} 0 \\ \ell\cos\theta \\ \ell\sin\theta \end{Bmatrix} = \begin{Bmatrix} -\ell\dot{\psi}\cos\theta \\ -\ell\dot{\theta}\sin\theta \\ \ell\dot{\theta}\cos\theta \end{Bmatrix}$$

$$\{\bar{a}_T(P)\}_B = \begin{Bmatrix} -\ell\ddot{\psi}\cos\theta + \ell\dot{\psi}\dot{\theta}\sin\theta \\ -\ell\ddot{\theta}\sin\theta - \ell\dot{\theta}^2\cos\theta \\ \ell\ddot{\theta}\cos\theta - \ell\dot{\theta}^2\sin\theta \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\psi} \end{Bmatrix} \times \begin{Bmatrix} -\ell\dot{\psi}\cos\theta \\ -\ell\dot{\theta}\sin\theta \\ \ell\dot{\theta}\cos\theta \end{Bmatrix} =$$

$$= \begin{Bmatrix} -\ell\ddot{\psi}\cos\theta + 2\ell\dot{\psi}\dot{\theta}\sin\theta \\ -\ell\ddot{\theta}\sin\theta - \ell\cos\theta\,(\dot{\theta}^2 + \dot{\psi}^2) \\ \ell\ddot{\theta}\cos\theta - \ell\dot{\theta}^2\sin\theta \end{Bmatrix}$$

$\bar{\alpha}_T^{brag} = \left.\dfrac{d\bar{\Omega}_T^{brag}}{dt}\right]_T$

$$\bar{\Omega}_T^{brag} = \bar{\Omega}_{forq}^{brag} + \bar{\Omega}_T^{forq} = (\odot\,\dot{\theta}) + (\Uparrow\,\dot{\psi})$$

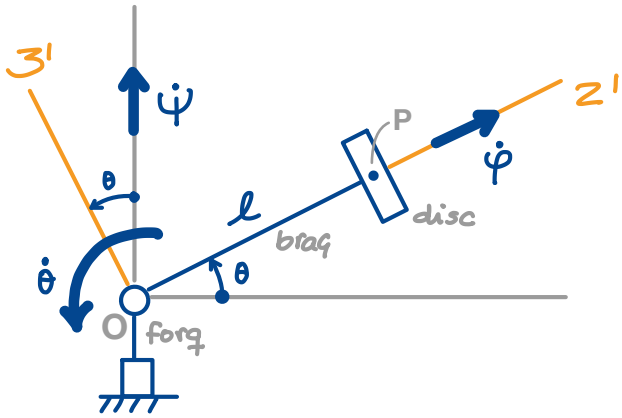
$$\{\bar{\Omega}_T^{brag}\}_B = \begin{Bmatrix} \dot{\theta} \\ 0 \\ \dot{\psi} \end{Bmatrix}$$

$\Big\downarrow$ derivant

$$\{\bar{\alpha}_T^{brag}\}_B = \begin{Bmatrix} \ddot{\theta} \\ 0 \\ \ddot{\psi} \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\psi} \end{Bmatrix} \times \begin{Bmatrix} \dot{\theta} \\ 0 \\ \dot{\psi} \end{Bmatrix} = \begin{Bmatrix} \ddot{\theta} \\ \dot{\psi}\dot{\theta} \\ \ddot{\psi} \end{Bmatrix}$$

En base B'

$\bar{v}_T(P)$ i $\bar{a}_T(P)$



$$\{\bar{OP}\}_{B'} = \begin{Bmatrix} 0 \\ \ell \\ 0 \end{Bmatrix}$$

$$\{\bar{v}_T(P)\}_{B'} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} + \underbrace{\begin{Bmatrix} \dot{\theta} \\ \dot{\psi}\sin\theta \\ \dot{\psi}\cos\theta \end{Bmatrix}}_{\bar{\Omega}_T^{B'}} \times \begin{Bmatrix} 0 \\ \ell \\ 0 \end{Bmatrix} = \begin{Bmatrix} -\ell\dot{\psi}\cos\theta \\ 0 \\ \ell\dot{\theta} \end{Bmatrix}$$

$$\boxed{\{\bar{\Omega}_T^{B'}\}_B = \underbrace{\begin{Bmatrix} \dot{\theta} \\ 0 \\ 0 \end{Bmatrix}}_{(\odot\dot{\theta})} + \underbrace{\begin{Bmatrix} 0 \\ \dot{\psi}\sin\theta \\ \dot{\psi}\cos\theta \end{Bmatrix}}_{(\Uparrow\dot{\psi})}}$$

$$\{\bar{a}_T(P)\}_{B'} = \begin{Bmatrix} -\ell\ddot{\psi}\cos\theta + \ell\dot{\psi}\dot{\theta}\sin\theta \\ 0 \\ \ell\ddot{\theta} \end{Bmatrix} + \begin{Bmatrix} \dot{\theta} \\ \dot{\psi}\sin\theta \\ \dot{\psi}\cos\theta \end{Bmatrix} \times \begin{Bmatrix} -\ell\dot{\psi}\cos\theta \\ 0 \\ \ell\dot{\theta} \end{Bmatrix} =$$

$$= \begin{Bmatrix} -\ell\ddot{\psi}\cos\theta + 2\ell\dot{\psi}\dot{\theta}\sin\theta \\ -\ell\dot{\psi}^2\cos^2\theta - \ell\dot{\theta}^2 \\ \ell\dot{\psi}^2\sin\theta\cos\theta + \ell\ddot{\theta} \end{Bmatrix}$$

$$\bar{\alpha}_T^{brac} = \left.\frac{d\bar{\Omega}_T^{brac}}{dt}\right]_T$$

$$\left\{\bar{\Omega}_T^{brac}\right\}_{B'} = \left\{\bar{\Omega}_T^{B'}\right\}_{B'} = \begin{Bmatrix} \dot{\theta} \\ \dot{\psi}\sin\theta \\ \dot{\psi}\cos\theta \end{Bmatrix}$$

Ja que B' és d'orientació fixa al brac

$$\left\{\bar{\alpha}_T^{brac}\right\}_{B'} = \begin{Bmatrix} \ddot{\theta} \\ \ddot{\psi}\sin\theta + \dot{\psi}\dot{\theta}\cos\theta \\ \ddot{\psi}\cos\theta - \dot{\psi}\dot{\theta}\sin\theta \end{Bmatrix} + \underbrace{\bar{\Omega}_T^{B'} \times \bar{\Omega}_T^{brac}}_{\bar{0}}$$

$$\bar{\Omega}_T^{B'} \parallel \bar{\Omega}_T^{brac}$$